Lecture 5 Exercise Solutions

1. An RSA cipher is set up with $\text{Key}_{\text{pub}} = (210757, 3)$ where 210757 is the modulus and 3 is the exponent. Encrypt $\text{Plaintext} = 12345$.

   $\text{Ciphertext} = 12345^3 \% 210757 = 187183$

   This is a fairly small number, so we really didn't need the Fast Exponentiation Algorithm.

2. An RSA cipher is set up with $\text{Key}_{\text{pub}} = (15943\ , 3)$, where 15943 is the modulus and 3 is the exponent. Find the decryption key.

   To find the encryption key from the decryption key requires us to factor 15943.

   $15943 = 107 \cdot 149$

   $p = 107$

   $q = 149$

   To find the d corresponding to $e = 3$, we need to find the inverse of 3 in $Z^{\times}_{15688}$. Why 15688? $\varphi(15943) = 15688$. Let's use the Extended Euclidean Algorithm to do that:

   $15688 = 5229 \cdot 3 + 1$

   $1 = (-5229) \cdot 3 + 1 \cdot 15688$

   So, the inverse of 3 in $Z^{\times}_{15688}$ is -5229. But, wait. -5229? Well, $Z^{\times}_{15688}$ is congruent to some value (block) in $Z^{\times}_{15688}$. It's easy to calculate: $-5229 \equiv k \cdot 15688 + -5229$. If $k = 1$, then $-5229 \equiv 15688 + -5229 = 10459$. So, $d = 10459$.

3. An RSA cipher is set up with $\text{Key}_{\text{pub}} = (12091\ , 3)$ modulus 12091 and the encryption key is 3. Decrypt $\text{Ciphertext} = 9812$.

   To decrypt the ciphertext we need the decryption key, but we don't have the decryption key—maybe we're the attacker, Eve. To be successful we need to do what we did in the previous problem: factor the modulus of the public key, calculate the modulus of the group, where the decryption key is the inverse of the encryption key. Then, given the decryption key raise the ciphertext to the power of the decryption key.

Factor:

$12091 = 107 \cdot 113$

$p = 107$

$q = 113$

Calculate modulus of the exponent group:

$\varphi(12091) = (p-1) \cdot (q-1) = 106 \cdot 112 = 11872$

Take the inverse:

$11872 = 3957 \cdot 3 + 1$

$1 = (-3957) \cdot 3 + 1 \cdot 11872$

$d = 7915 \equiv -3957 \in Z_{11872}^{\times}$

Decrypt $\text{Plaintext} = 9812^{7915} \% 11872$:

X	e	Y
9812	7915	1
9812	7914	9812
6802	3957	9812
6802	3956	10995
7038	1978	10995
8708	989	10995
8708	988	7922
6603	494	7922
11544	247	7922
11544	246	1918
10276	123	1918
10276	122	1038
5473	61	1038
5473	60	10295
4322	30	10295
11180	15	10295
11180	14	3871
7733	7	3871
7733	6	9218
9294	3	9218
9294	2	7357
332	1	7357
332	0	142

So, Plaintext = $9812^{7915} \% 11872 = 142$.